

Two kinds of AI in research math

A Millennium Prize claim, and an afternoon with a coding assistant

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Why this talk

- On 8 September, OpenAI said its AI had settled a 90-year-old, \$1M question about how fluids move.
- The night before, a mathematician and a coding assistant went public with a year of closely related work.
- Two weeks ago I rebuilt the simulation methods from my 2016 dissertation with Claude Code, in a day.

Thesis

“AI in research math” currently means two very different things. Both are worth seeing up close.

Part 1 (15 min). The Navier-Stokes claim: what was claimed, how it was produced, who else was there, what is still open.

Part 2 (15 min). The other kind: wave simulations from my dissertation, re-derived with an assistant. Two short videos.

Part 1: the Navier-Stokes claim

Fluids, Newton, and a question from 1934

- The **Navier-Stokes equations** are Newton's $F = ma$ applied to a continuous fluid: water, air, blood. They sit inside aircraft design, weather forecasting, and ocean models.
- Inputs: how the fluid is moving at the start, plus an optional “stirring” force. Output: velocity and pressure everywhere, forever.
- The open question (Leray, 1934): if the fluid starts out smooth, do the equations always keep predicting smooth motion, or can they predict **infinite speed in finite time** (“blowup”)?
- Real fluids cannot move infinitely fast. Blowup would mean the model leaves its own range of validity and molecules have to take over.
- In 2000 the Clay Mathematics Institute offered \$1M for an answer, either way.

The equation itself is in the backup slides. Nothing today depends on reading it.

Four ways to win the prize (Fefferman, 2000)

	Show that...	Where	Stirring force?
(A)	smooth solutions always exist	all of space	none
(B)	smooth solutions always exist	a periodic box	none
(C)	one smooth start has no smooth solution	all of space	smooth, allowed
(D)	one smooth start has no smooth solution	a periodic box	smooth, allowed

- (A)/(B) would be the reassuring answer; (C)/(D) would be a counterexample.
- The official statement lets a counterexample use a smooth, well-behaved external force. The picture most people carry, “can turbulence by itself blow up?”, is the *unforced* question. Keep that distinction in mind.

Source: C. Fefferman, *Existence and smoothness of the Navier-Stokes equation*, Clay Mathematics Institute, p. 2.

What happened, in two weeks

- Aug 15-22** Alpöge and Buckmaster get blowup with smooth forcing for two model equations, including 3-D Euler; Lean-verified a week later.
- Aug 28** OpenAI starts training a new internal model.
- Sep 1** Rumours that “two Millennium Prize problems had been resolved”; OpenAI points agent groups at all six open problems.
- Sep 3** Buckmaster emails OpenAI.
- Sep 5** OpenAI’s agents reach the Navier-Stokes result, 88 hours after launch.
- Sep 6** Lean formalization done (17 h). Two calls between Buckmaster and OpenAI.
- Sep 7, night** Buckmaster and Alpöge post three papers, Lean code, and a statement. A Caltech group posts a third, numerical route.
- Sep 8** OpenAI announces: blog post, 166-page manuscript, 57-page Euler companion, Lean repository.
- Sep 11** Clay: the problem “has apparently been settled”; review “deliberately unhurried”. Twenty-five Fields Medalists publish *A Severe Misalignment of AI in Mathematics*.
- Sep 16-17** Nature editorial on attribution. Clay’s site lists the problem as “Active”, not “Solved”.

Sources: OpenAI’s post; Buckmaster’s statement; claymath.org; mathandai.org; nature.com. Every date is traced in docs/navier-stokes-notes.md.

The claim, in words

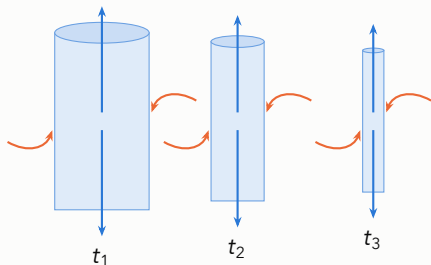
Abstract, verbatim

"For every positive viscosity, we construct a solution of the three-dimensional incompressible Navier-Stokes equations that starts from rest and develops unbounded velocity in finite time while maintaining uniformly bounded kinetic energy."

- The fluid starts **at rest**. A smooth, finite stirring force is applied; it is part of the construction, not something you would meet in a river.
- Total kinetic energy stays bounded the whole time.
- Yet the top speed goes to infinity as $t \rightarrow 1$. So statements (C) and (D) hold: for this force and start, no smooth solution exists for all time.
- **What it is not.** The unforced questions (A)/(B) are untouched. Nobody has shown that turbulence on its own blows up.

Source: OpenAI, *Finite Time Blowup for Navier-Stokes*, Theorem 1.1 (p. 1) and Corollary 10.6 (p. 117).

The picture: a collapsing vortex



After Fig. 1 of the manuscript. Orange: inward spiral; blue: outflow along the axis.

- A spinning column whose radius shrinks faster than its height: thinner, longer, faster. OpenAI's phrase: "like spaghetti".
- Fluid spirals in and leaves along the axis. Pulling mass inward speeds up the spin (the figure-skater effect). Viscosity pushes back and loses.
- Energy stays finite because the fast region shrinks faster than the speed grows.
- The catch: the force needed to hold the bare vortex together would itself blow up. The fix: a sequence of fine **ripples** whose own momentum transport cancels the bad part. What is left is a smooth force.

How it was produced, in OpenAI's own account

- An internal model “significantly more capable than GPT-6 Astra”, in training since 28 August.
- A system of **coordinating agents** with tools (a cached copy of the internet, code execution), split into groups that talk within the group. Different groups were given the four variants (A)–(D).
- Warm-up: the unforced Euler question fell first, to “nearly 100 agents” in about 50 hours. Everything was then pointed at Navier–Stokes, seeded with that solution; Codex was used to cross-pollinate the groups.
- “On the order of 10,000 concurrent agents”, 88 hours, 2.7 million messages, about 130 billion output tokens. Lean formalization: 17 more hours with GPT-6 Astra.
- Cost: “millions of dollars” (Mark Chen, to *Science*). Outside estimates run from \$1M to tens of millions. OpenAI will not claim the prize.

Source: openai.com/index/navier-stokes-solution (8 Sep, updated 10 Sep); *Science*, 8 Sep. *Nature* and *Science* quote the briefing as “1,000 agents” for the warm-up; the post says “nearly 100”. A small reminder to check press numbers against the primary source.

The parallel story: a year, two people, several models

- Tristan Buckmaster (NYU) and Levent Alpöge (Anthropic), a personal collaboration paid from Buckmaster's own research funds.
- The programme: forced-blowup constructions of Córdoba and Martínez-Zoroa (Madrid). Goal: push them to smooth forcing and to 3-D Euler. Tools: Claude and Codex (GPT-5.6 Sol; Astra for write-ups and auditing).
- Slow for most of a year; results on 15 August, Lean-verified 22 August. "The first LLM generated proof Levent sent me was the most horrendous I have ever read."
- Released 7 September, earlier than planned: "The Euler writeup, in particular, can only be described as AI slop."
- His framing: "This is a Deep Blue-Kasparov moment."
- Same night, a third route: Anandkumar's group (Caltech) posted a numerically stable unforced-Euler blowup candidate found with a physics-informed neural network. Not yet a proof.

Sources: Buckmaster's statement (cims.nyu.edu/~tristanb); Tao, 8 Sep; anima-ai.org, 7 Sep. Fefferman, to Quanta: "The heroes of the story are Córdoba and Martínez-Zoroa."

The dispute, as each side tells it

Buckmaster's statement (7 Sep)

- Emailed OpenAI on 3 Sep; two calls on 6 Sep with Bubeck and an OpenAI mathematician. Told the proof came with "very little human input", which "turned out not to be true".
- "I asked whether the model had been trained on, or had access to, our sessions in Codex ... I asked again, about training, and I did not get an answer."
- Two proposals, one a paper by Buckmaster alone crediting an OpenAI model, with Alpöge removed. Declined both. Then: "If you don't want me to be nice, then I don't have to be nice."
- "I am not accusing anyone of anything. I am stating what I was told, when, and what was proposed to me."

OpenAI's post (8 Sep, updated 10 Sep)

- The effort "began on September 1st after hearing a rumor which we later realized was related to" the pair.
- "We (the researchers and the agents) did not see any of their work through any means until they released it publicly."
- 8 Sep: "we cannot rule out that de-identified data derived from their usage of our products helped improve our models." 10 Sep, after an investigation: their Codex prompts "could not have influenced the system in any way, including through training."
- "We recognize the priority of their work on forced Euler"; offered to show the prompts and the proof.

Neither account is independently verified. This slide reports claims and responses, not findings.

Reactions

- **Clay Mathematics Institute** (11 Sep): “apparently been settled”; evaluation and credit “deliberately unhurried”. Site status: *Active*.
- **American Mathematical Society** (8 Sep): “a milestone advance in human knowledge”.
- **25 Fields Medalists** (11 Sep): “solving problems is only a tool and proxy for achieving the primary goal of conceptual understanding and insight”; rushed announcements raise “severe attribution and plagiarism questions”.
- **Terence Tao**: open problems “are being mined as a non-renewable resource”; “even the rumor of someone working on a problem can trigger a massive amount of AI-powered effort to flatten it”.
- **Michael Harris** (Columbia, to *Science*): it “convinces decision makers that human mathematicians are obsolete”.
- **Sébastien Bubeck** (OpenAI): “the spectacular culmination of the arc we have seen over the last 12 months”; next, “developing new materials, finding cures to diseases”.
- **Nature editorial** (16 Sep): AI companies must work with the research community to protect attribution.

What “machine-checked” does and does not mean

Lean checks

- That every step follows from three standard axioms. No gaps, no “clearly”.
- Reproducibly: clone the repository, build it, run the independent checker.

Humans still check

- That the formal theorem *is* the Clay statement. OpenAI targeted an independent transcription of it (Google DeepMind’s), which helps. Still a judgment.
- Whether the construction teaches anything.
- Clay: publication, a two-year wait, then a committee.

Quanta, 8 Sep

“The crucial bit of verification that must still be done by humans is to guarantee that the statement being shown to be true in Lean is logically equivalent to what mathematicians set out to prove.”

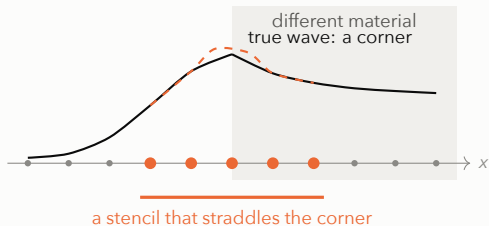
Questions to argue about

1. **Letter vs. spirit.** The statement allows a smooth force, and this result uses one. Is the Millennium problem solved?
2. **Credit.** Córdoba and Martínez-Zoroa's ideas; Alpöge and Buckmaster's year; OpenAI's 88 hours. Who gets what?
3. **Trust.** Would you put unpublished work into a frontier lab's tools? What would you need to see first?
4. **Compute.** No university can run 10,000 agents for four days. What does that do to a field whose scarce resource was always ideas?
5. **What is still human?** Choosing the problem, the physical picture, judging what matters, deciding what to formalize, and writing it so other people understand it.

Part 2: the other kind

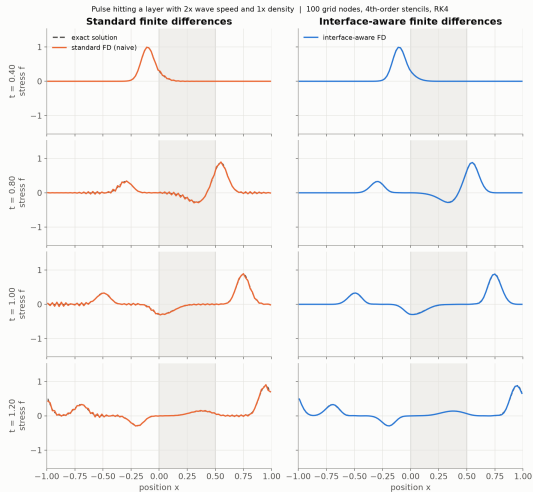
- 2016 PhD, applied mathematics, CU Boulder: simulating waves travelling through **layered materials**. Seismic imaging (sound through rock), ultrasound (sound through tissue).
- The contribution: “interface-aware” stencils, that is, simulation building blocks that know where the material changes.
- Written in MATLAB, nine years ago. I no longer have a licence, and the code was never meant for anyone else to read.
- The experiment: with Claude Code, re-derive the method from the papers, port it to Python, and verify it against exact solutions.
- Result: the 1-D method and the 2-D method, both working and tested, in one day.

The problem in one picture



- A simulation stores the wave at grid points and estimates slopes from a few neighbours: a **stencil**.
- Where the material changes, the wave is continuous but its *slope jumps*: a corner.
- A stencil straddling the corner assumes a smooth wave (dashed) and gets the slope wrong. The error shows up as ringing and wrong reflections.
- The fix: rebuild the handful of stencils near the boundary from pieces that obey the jump conditions. Everything else is unchanged.

1-D: the ringing, and the fix

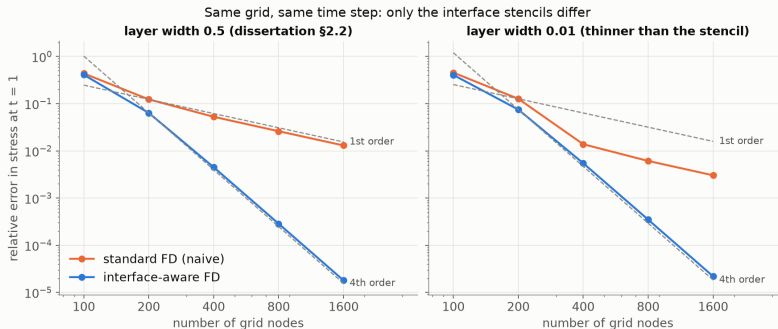


- A pulse hits a layer (shaded) where waves travel twice as fast. Part reflects, part transmits.
- Both panels: 100 grid points, the same time step, the same code. Only the stencils touching the layer edges differ.
- **Standard**: a trail of wiggles behind every reflection; 10% error by $t = 1$.
Interface-aware: on the exact solution; 3% error, all of it from the coarse grid.

Play clip:

wave1d_naive_vs_aware_coarse.mp4 (10s)

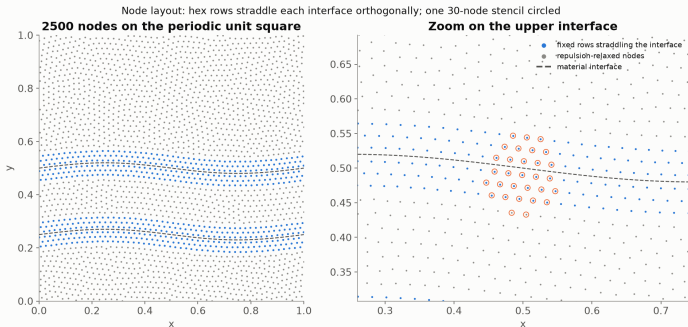
1-D: how fast the error shrinks



- **Standard**: double the points, halve the error ("first order"). The corner caps the accuracy however good the stencils are elsewhere.
- **Interface-aware**: double the points, divide the error by 16 ("fourth order"). Also for a layer thinner than the stencil (right).

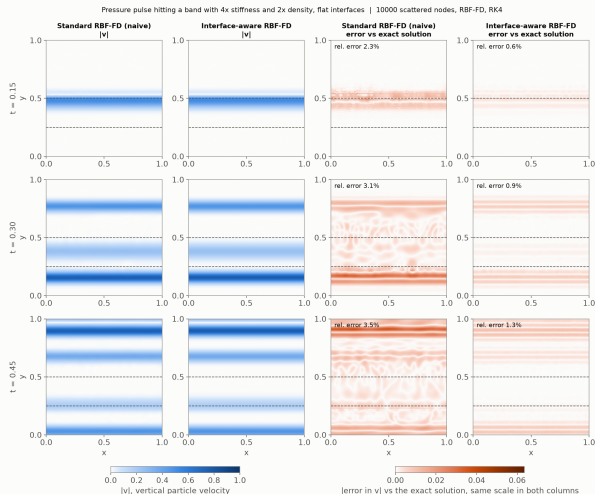
Reproduces Fig. 2-8 of the dissertation. Error measured against an exact ray-sum solution at $t = 1$.

2-D: scattered points and curved boundaries



- No grid. Points are spread by a repulsion process, with fixed rows straddling each boundary (blue). That layout was the key to stability; it took years to find and is still unproven.
- Each point's stencil is its 30 nearest neighbours, weighted with radial basis functions plus polynomials (RBF-FD). Five fields: two velocities, three stresses.

2-D: same points, only the stencils near the boundaries differ

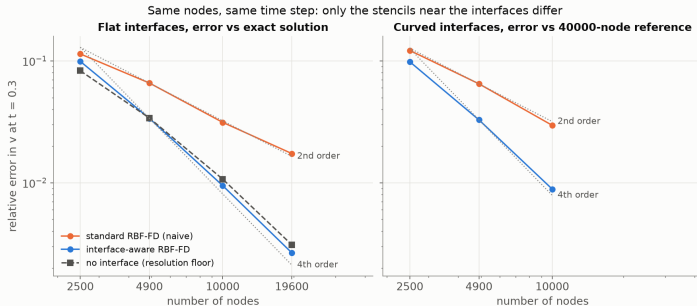


- A pressure pulse travels down through a band of stiffer, denser material. 10,000 points.
- Left pair: the wave. Right pair: the error against the exact solution, on one colour scale.
- At $t = 0.3$: **standard** 3.1% error, **interface-aware** 0.9%. The standard error starts at the boundary, then travels with the wave.

Play clip:

wave2d_naive_vs_aware.mp4, then
_curved.mp4 (8 s each)

2-D: the boundary error is gone



- Dashed: the same pulse in a uniform material, so pure resolution error with no boundary at all.
- The **interface-aware** error sits on that floor: at these resolutions there is nothing left to fix at the boundary. Curved boundaries (right) give the same numbers.

Reproduces Fig. 3-5 and 3-8 of the dissertation. Left: error against the exact solution. Right: against a 40,000-point reference run.

What the collaboration looked like

One day (17 Sep)

- 8 issues, 7 pull requests, 26 commits, 106 tests, all green.
- Every numerical routine tested against an exact solution or a convergence rate, not "it runs".
- Independent reference solutions built before trusting a solver.
- Adversarial review passes caught three real bugs: a sign error, two latent bugs in the exact solver that only the 2-D problem exposed, a crash in a driver.

What was still human

- Where to put the points (the straddling rows).
- Plain method first, clever method second.
- Which basis functions, and the narrow window for the stabilising parameter.
- "Don't let the error measurement itself cross a boundary."
- Same list as Part 1: steering, judging what matters, designing the verification. The typing, the bookkeeping, and the port were not.

Two kinds of AI in research math

	OpenAI, Navier-Stokes	Alpöge and Buckmaster	This afternoon
Who steered	a lab team, via prompts	two mathematicians	one engineer
Scale	~10,000 agents, 88 h	about a year, several models	one assistant, one day
Verification	Lean; Clay review pending	Lean; papers being rewritten	tests against exact solutions
Who understands it	contested	"AI slop", being digested	fully; it is my method
Cost	millions of dollars	own research funds	a subscription

- The capability is real at both ends of this table.
- The difference is who steered, who can check it, and who is left understanding the result.

For the curious

Gentle starts

- 3Blue1Brown, [But what is a partial differential equation?](#) (YouTube, 17 min)
- Khan Academy: [Differential equations](#); [Introduction to partial derivatives](#)
- Wikipedia: [Navier-Stokes existence and smoothness](#); [Finite difference method](#); [Lean](#) (proof assistant)

Primary sources for Part 1

- OpenAI: [the announcement and manuscript](#); [the Lean repository](#)
- Clay Mathematics Institute: [the problem page and official statement](#)
- [Buckmaster's statement](#) and papers (cims.nyu.edu/~tristanb)
- [Tao's blog post, 7 Sep 2026](#); [the Fields Medalists' declaration](#)

Part 2

- Code, tests, figures, and these notes: github.com/bradleypmartin/20260930-zd-ai-pdes-demo (MIT)

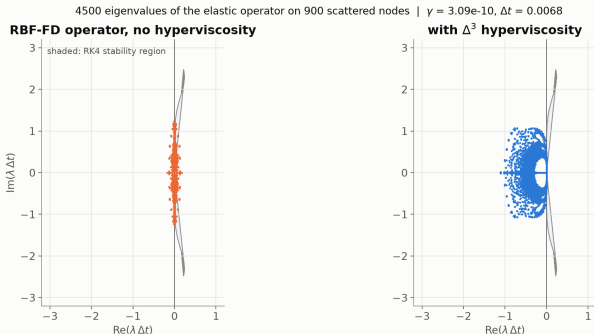
Backup: the equations

For velocity $u(x, t)$, pressure $p(x, t)$, viscosity $\nu > 0$, and force f :

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0.$$

- Left to right: acceleration, transport by the flow itself (the nonlinear term, and the source of all the difficulty), viscous smoothing, pressure. Right: the stirring force.
- Second equation: the fluid is incompressible.
- Theorem 1.1: for every $\nu > 0$ there are a smooth compactly supported f and smooth (u, p) on $\mathbb{R}^3 \times [0, 1)$ with $u(\cdot, 0) = 0$, $\sup_t \|u(t)\|_{L^2} < \infty$, and $\limsup_{t \uparrow 1} \|u(t)\|_{L^\infty} = \infty$.

Backup: keeping the 2-D method stable



- Scattered-point stencils leave a few growing modes (left, orange outside the stability region). A small high-order damping term (“hyperviscosity”) moves them inside (right) without touching the accurate ones.
- Its size has to scale with the point spacing in a narrow window. That window was a human input; the eigenvalue check was the assistant’s.

Backup: Part 2 numbers

1-D, relative error in stress at $t = 1$

points	standard	interface-aware
200	1.2×10^{-1}	6.4×10^{-2}
400	5.3×10^{-2}	4.5×10^{-3}
800	2.6×10^{-2}	2.9×10^{-4}
1600	1.3×10^{-2}	1.8×10^{-5}

2-D, flat boundaries, error in v at $t = 0.3$

points	standard	aware	no boundary
2500	1.1×10^{-1}	9.9×10^{-2}	8.2×10^{-2}
4900	6.9×10^{-2}	3.4×10^{-2}	3.3×10^{-2}
10000	3.1×10^{-2}	9.6×10^{-3}	1.1×10^{-2}
19600	1.7×10^{-2}	2.7×10^{-3}	3.0×10^{-3}

From docs/demo-outline.md; drivers scripts/wave1d_convergence.py and scripts/wave2d_convergence.py.